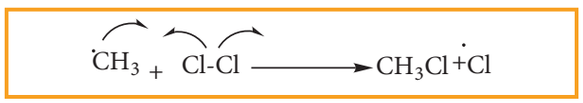


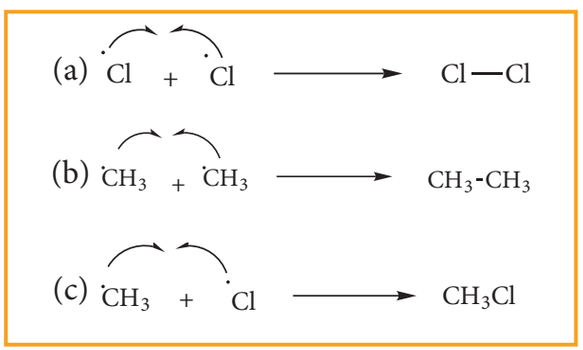
(b) The methyl free radical thus obtained attacks the second molecule of chlorine to give chloromethane (CH_3Cl) and a chlorine free radical as follows.



(c) This chlorine free radical then cycles back to step (a) and both step (a) and (b) are repeated many times and thus chain of reaction is set up.

iii) Chain termination:

After sometimes, the reactions stops due to consumption of reactant and the chain is terminated by the combination of free radicals.



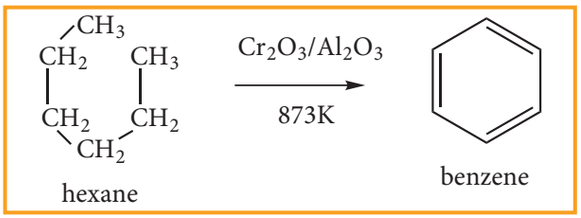
Evaluate Yourself

7) Why ethane is produced in chlorination of methane?

3) Aromatisation

Alkanes with six to ten carbon atoms are converted into homologous of benzene at high temperature and in the presence of catalyst. This process is known as aromatization. It occurs by simultaneous cyclisation followed by dehydrogenation of alkanes.

n-Hexane passed over Cr_2O_3 supported on alumina at 873 K gives benzene.

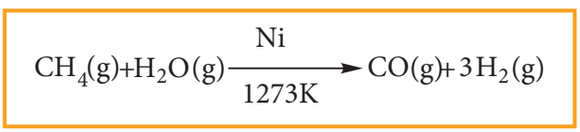


Evaluate Yourself

8) How toluene can be prepared by this method?

4) Reaction With Steam:

Methane reacts with steam at 1273K in the presence of Nickel and decomposes to form carbon monoxide and hydrogen gas.



Production of H_2 gas from methane is known as steam reforming process and it is a well-established industrial process for the production of H_2 gas from hydrocarbons.

5) Pyrolysis

Pyrolysis is defined as the thermal decomposition of organic compound into smaller fragments in the absence of air through the application of heat. 'Pyro' means 'fire' and 'lysis' means 'separating'. Pyrolysis of alkanes also named as cracking.

In the absence of air, when alkane vapours are passed through red-hot metal it breaks down into simpler hydrocarbons.

Unit 13 - jagan.indd 189 22-01-2021 12:51:30